Homework 2: Cosmology

Solving the Friedmann Equation

The Friedmann Equation for a matter dominated universe in non-dimensionalized form is

$$(\tilde{R}')^2 - 2\tilde{R} = -k\tilde{R}^2 \tag{1}$$

where $\tilde{R} = R/R^*$, and the prime denotes derivation with respect to the non-conformal time $d\eta = \frac{c\,dt}{R}$.

1. Solve the matter-dominated Friedmann Equation for all three cases (closed, flat, open) to obtain an expression for $R(\eta)$ and $t(\eta)$.

   If $k = 0$ and $\tilde{R}(0) = 0$, then we obtain:

$$\begin{aligned} \sqrt{(\tilde{R}')^2} &= \sqrt{2\tilde{R}} && \text{by addition} \\ \frac{1}{\sqrt{2}}\frac{\tilde{R}^{1/2}}{1/2} &= \int \frac{d\tilde{R}}{\sqrt{2\tilde{R}}} = \int d\eta = \eta + 0 && \text{by integration} \\ \tilde{R}(\eta) &= \frac{\eta^2}{2} \end{aligned}$$

   Otherwise, the Friedmann Equation with initial value $\tilde{R}(0) = 0$ yields:

$$\begin{aligned} \sqrt{\left(\tilde{R}'\right)^2} &= \sqrt{-k\tilde{R}^2 + 2\tilde{R} - \frac{1}{k} + \frac{1}{k}} \\ \int \frac{k\; d\tilde{R}}{\sqrt{1-(k\tilde{R}-1)^2}} &= \int \frac{k}{\sqrt{k}}\, d\eta \end{aligned}$$

$$\tilde{R} = \frac{1}{k}\left(\sin\left(\sin^{-1}(k\tilde{R}-1)\right)+1\right) = \frac{\sin\left(\eta\sqrt{k}+C\right)+1}{k},$$

   where $C = \sin^{-1}\left(\dfrac{\sin\left(0\sqrt{k}+C\right)+1}{k}\cdot k - 1\right) = \sin^{-1}(0\cdot k - 1) = -\dfrac{\pi}{2}$ thus yields:

$$\begin{aligned} \tilde{R}(\eta) &= \frac{1}{k}\left(\sin(\eta\sqrt{k}-\frac{\pi}{2})+1\right) = \frac{1}{k}\left(\cos(\eta\sqrt{k})+1\right) = \frac{2}{k}\sin^2\left(\frac{\eta\sqrt{k}}{2}\right) \\ &= \frac{2k}{k}S_k^2\left(\frac{\eta}{2}\right) = \begin{cases} 2\sin^2\left(\frac{\eta}{2}\right) & \text{if } k=+1 \\ 2\sinh^2\left(\frac{\eta}{2}\right) & \text{if } k=-1 \end{cases}. \end{aligned}$$

In all three cases, we obtain:

$$R(\eta) = \tilde{R}\frac{G}{c^2}\rho\frac{4}{3}\pi R^3 = \sqrt{\frac{3c^2}{4G\rho\pi\tilde{R}(\eta)}}$$

If $\eta(0) = 0$, then we obtain

$$t(\eta) = \int dt = \int \frac{R\ d\eta}{c}.$$

2. Use your favorite plotting program to plot the non-dimensionalized scale factor R/R^* versus non-dimensionalized time $ct/(2\pi R^*)$.

See Jupyter notebook attached.
The general Friedmann equation can be solved numerically with a variety of assumptions about matter, radiation, and vacuum density.

3. Show that the Friedmann Equation can be written in non-dimensionalized form: $\left(\frac{da}{d\tilde{t}}\right)^2 = \Omega_m a^{-1} + \Omega_r a^{-2} + \Omega_v a^2 - (\Omega_0 - 1)$, where $\tilde{t} = H_0 t$ and $a = R/R_0$. We obtain the result as follows:

$$\Omega_v + \Omega_m a^{-3} + \Omega_r a^{-4} - (\Omega_0 - 1)a^{-2} = \left(\frac{H}{H_0}\right)^2 = \left(\frac{1/R_0}{RH_0/R_0}\frac{dR}{dt}\right)^2$$

$$a^{-2}\left(\Omega_m a^{-1} + \Omega_v a^2 + \Omega_r a^{-2} - (\Omega_0 - 1)a^0\right) = \left(\frac{1}{R/R_0}\frac{d(R/R_0)}{d(H_0 t)}\right)^2$$

$$\left(\frac{1}{a}\right)^2\left(\Omega_m a^{-1} + \Omega_r a^{-2} + \Omega_v a^2 - (\Omega_0 - 1)a^0\right) = \left(\frac{1}{a}\right)^2\left(\frac{da}{d\tilde{t}}\right)^2$$

$$\Omega_m a^{-1} + \Omega_r a^{-2} + \Omega_v a^2 - (\Omega_0 - 1) = \left(\frac{da}{d\tilde{t}}\right)^2$$

4. Use an ODE solver to solve this equation numerically. Choose intial conditions of $a(0) = 0.1$ to avoid an initial singularity. Plot solutions for a flat universe ($\Omega_0 = 1$), with $\Omega_r = 0.4$, $\Omega_m = 0.4$, and $\Omega_v = 0.2$. Plot the results and identify periods of a radiation-dominated, matter-dominated, and vacuum-dominated universe.

   When $0 \leq t \ll 0.7$, we have $a(t) \approx t$ and thus $a^2 < a < a^{-1} < a^{-2}$ implies that the universe is radiation-dominated. At $t = 0.7$, there is a rough equilibrium between radiation and matter. When $0.7 \leq t \leq \sqrt[3]{2} \approx 1.25$, the universe is matter-dominated. When $1.25 \leq t$, the universe is vacuum-dominated.

   Indeed, our figures show that the $a(t)$ curve looks like $t^{1/2}$, $t^{2/3}$, and e^t on those three sections.